Learning to Hedge from Simulated Experience

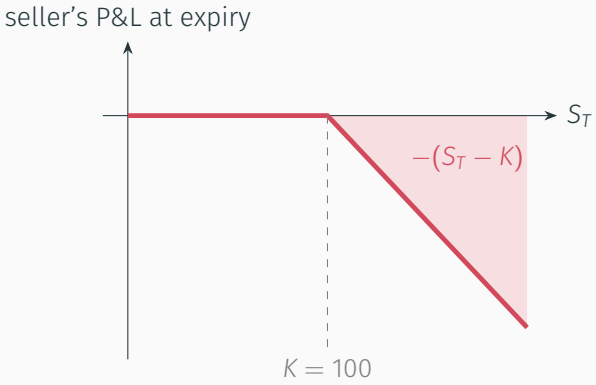
Deep Reinforcement Learning for Option Hedging with Transaction Costs

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The problem: a dealer sold a call



One-month call, strike 100. Premium fixed, payout unbounded.

Each day: how many shares to hold?

Why textbook delta is not enough

Black–Scholes delta replicates the call exactly, if:

- trading is continuous
- trading is free
- volatility is known

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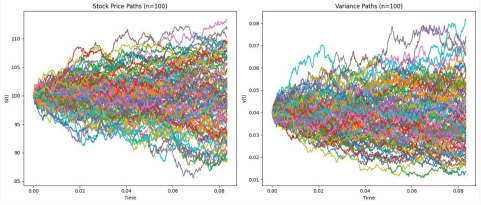
Black–Scholes delta replicates the call exactly, if:

- trading is continuous
  we trade once a day
- trading is free
  each trade costs 0.1%
- volatility is known

Why textbook delta is not enough

Black–Scholes delta replicates the call exactly, if:

- trading is continuous
  we trade once a day
- trading is free
  each trade costs 0.1%
- volatility is known
  here it is random



No perfect hedge exists:
choose which risk to keep, at what cost.

Two ways to get a hedge

Derive a rule

Model the market, solve for the hedge.

Interpretable, but each new friction needs a new derivation.

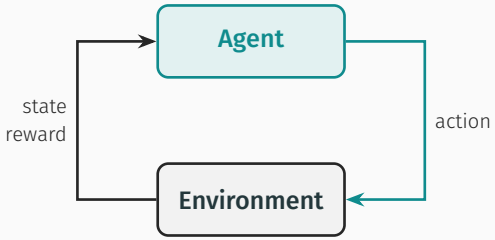
Learn a policy

Simulate the market, learn by trial.

Frictions are just rules of the simulator, but learning is data-hungry.

Both need a market model.

Reinforcement learning in one picture



The agent learns a policy that maximises expected reward, by trial, from samples of the environment.

Hedging as a game

State

Market + book

price, variance, position, time, P&L

Hedging as a game

State

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price, variance, position, time, P&L

Action

Hedge position

shares to hold, $h \in [0, 1]$

Hedging as a game

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shares to hold, $h \in [0, 1]$

Reward

Hedging P&L

terminal + mark-to-market

Hedging as a game

State	Action	Reward	Episode
Market + book	**Hedge position**	**Hedging P&L**	**Option life**
price, variance, position, time, P&L	shares to hold, $h \in [0, 1]$	terminal + mark-to-market	1 month, 22 trading dates

Costs and position limits are rules of the game.

The learners: TD3 and SAC

TD3

Twin critics, cautious value estimates.

SAC

Entropy bonus, keeps exploring.

Both are actor–critic methods for continuous actions.

Off-the-shelf: Stable-Baselines3, default settings.

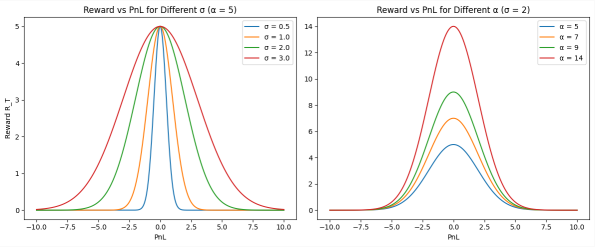
The reward

At expiry:

$$R_T = \alpha \exp\Big(- \frac{\mathrm{PnL}_T^2}{2\sigma^2} \Big)$$

Rewards ending near zero; bounded.

Plus a daily mark-to-market signal.



Experiment

Market
Heston, $\rho = -0.7$

Liability
ATM call, 1 month

Trading
22 daily dates, 0.1% cost

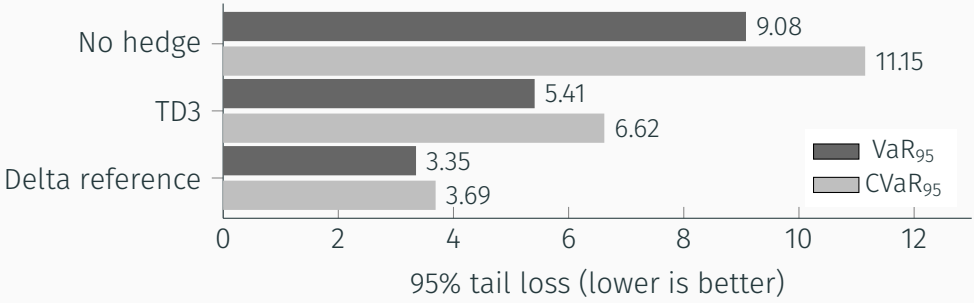
Training
40 000 steps

Test
10 000 new paths

Tail risk
VaR and CVaR at 95%

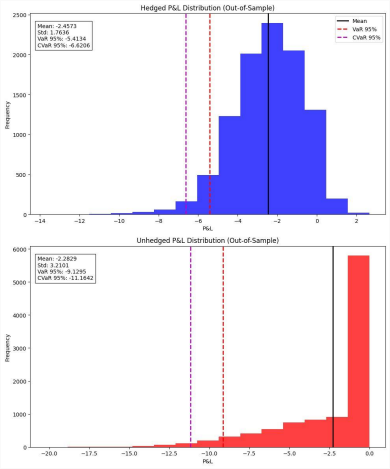
Reference: Black–Scholes delta, given the true volatility.

Result: most of the way to the textbook hedge



TD3 recovers ≈ 60% of the tail-risk reduction of delta, from 40 000 training steps.

What the P&L looks like

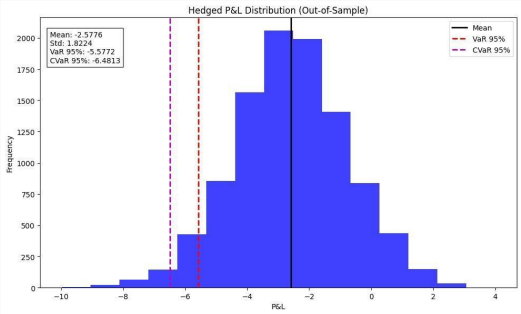


	Mean	Std	VaR_{95}	CVaR_{95}
No hedge	−2.31	3.22	9.08	11.15
TD3	−2.46	1.76	5.41	6.62
Delta reference	−2.49	0.50	3.35	3.69

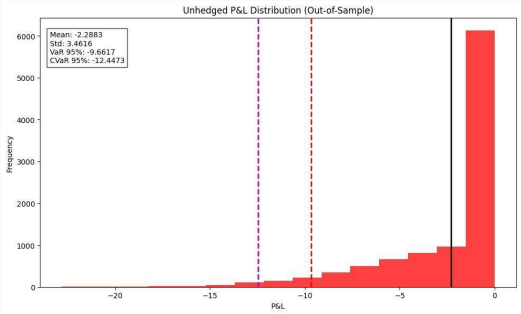
Top: TD3. Bottom: no hedge.

The hedge pulls in the left tail; delta narrows it further.

Same pipeline, different market



TD3 hedged



No hedge

Retrained on a Black–Scholes simulator: VaR_{95} −42%, CVaR_{95} −48% vs no hedge.

Takeaway

A generic learner, with no hedging rule built in,
recovers most of a textbook hedge.

Next: train on one model, test on another
cost-aware benchmarks · several seeds

Thank you. Questions?

`github.com/platonbarunin/diploma`

Backup

Backup: How the delta reference is calculated

1. 10 000 Heston paths, the paper's parameters (seed 0, 400 steps, one month).
2. On each of 22 daily dates hold $N(d_1)$ shares with $\sigma = \sqrt{v_t}$, the true volatility the agent also sees.
3. Agent's rules: 0.1% cost, $h \in [0, 1]$, zero capital, premium not booked.
4. At expiry: $\mathrm{PnL} = h\,S_T + \text{cash} - (S_T - K)^+$.

	Mean	Std	VaR_{95}	CVaR_{95}
No hedge	−2.31	3.22	9.08	11.15
TD3 (RL)	−2.46	1.76	5.41	6.62
SAC (RL)	−2.46	1.87	5.75	6.92
BS delta ($\sqrt{v_t}$)	−2.49	0.50	3.35	3.69

No hedge and delta: own seeded paths. TD3, SAC: the paper's test run, a different set of 10 000 paths from the same simulator. TD3 closes 64% (VaR) and 61% (CVaR) of the gap to delta. Code: `baseline/`.

Backup: what the comparison shows, and what it does not

Shows

- a generic actor–critic with default settings and $\approx 2\,200$ decisions of experience removes $\approx 40\%$ of tail loss under costs and stochastic volatility
- it recovers most, not all, of what a good model-based rule achieves
- the reference is strong: true instantaneous variance, and a short ATM option for which the Black–Scholes form is a good approximation

Does not show

- an advantage over model-based hedging
- robustness to model error: training and test use the same simulator
- stability across runs: these are the best configurations of an (α, σ) grid
- performance on real market data

Backup: reward caveats

Terminal reward $R_T = \alpha e^{-\mathrm{PnL}_T^2/2\sigma^2}$

- not a tail measure: at $\sigma = 0.5$ a loss of 3 and a loss of 10 both score ≈ 0
- target 0 is off-centre: the premium is not booked, so every strategy has mean P&L $\approx -(\text{price} + \text{costs})$, between -2.3 and -2.5

Mark-to-market $R_t = \widehat{\mathrm{PnL}}_t - \widehat{\mathrm{PnL}}_{t-1}$, $\widehat{\mathrm{PnL}}_t = V_t(H) - \hat{C}_t$

- $\hat{C}_t$ extends the ATM approximation $0.4\, S_t\sqrt{v_t \tau}$ (Stefanica 2011): some model structure enters through the reward
- the shaping is not potential-based, so it can shift the optimal policy (Ng et al. 1999)

Discounting: $\gamma = 0.99$ per simulator step over 400 steps, so the terminal reward is weighted $0.99^{399} \approx 0.02$ at the first decision.

Fixes: premium-inclusive P&L, a CVaR-type objective, $\gamma = 1$ or 22-step episodes.

Backup: why the market is incomplete

Heston under $\mathbb{Q}$:

$$dS_t = rS_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S, \qquad dv_t = \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dW_t^v, \qquad d\langle W^S, W^v\rangle_t = \rho\,dt.$$

Write $dW^v = \rho\,dW^S + \sqrt{1-\rho^2}\,dW^\perp$.

- **One** traded risky asset, **two** Brownian motions.
- Trading the stock offsets dW^S (and the ρ-part of volatility risk); the $\sqrt{1-\rho^2}\,dW^\perp$ part of the option's vega exposure cannot be offset.
- Hence many martingale measures (indexed by the market price of volatility risk): no unique price, no perfect hedge, even in continuous time without costs.
- Completing it: trade a second option or a variance swap.

Costs alone also break replication: with proportional costs the cheapest super-replicating strategy for a call is to buy one share (cost S_0) (Soner, Shreve & Cvitanić 1995). In practice one trades off risk against cost: exactly what the reward encodes.

Backup: the formulation in full

- **General form (paper)**: POMDP $(\mathcal{S}, \mathcal{A}, \Omega, \mathcal{T}, \mathcal{O}, \mathcal{R})$; latent state $\mathcal{S}$ = true factors driving prices, simulated by Heston.
- **Action** $a_t \in \mathcal{A} = [0, 1]$: new number of shares h_{t+1}; trade $\Delta h = h_{t+1} - h_t$ pays $C_t = \lambda \, |\Delta h| \, S_t$, $\lambda = 0.001$.
- **Observation** $o_t = (\tilde{S}_t, \tilde{v}_t, h_t, \tau_t, \widetilde{\mathrm{PnL}}_t, m_t)$, all but the moneyness flag m_t scaled to a bounded interval; $\mathcal{F}_t$-adapted, no look-ahead.
- **Transition**: Euler–Maruyama Heston; no market impact. $\kappa, \theta, \sigma, \rho$ unknown to the agent.
- **Reward**: $R_t = \widehat{\mathrm{PnL}}_t - \widehat{\mathrm{PnL}}_{t-1}$ for $t < T$, plus $\alpha e^{-\mathrm{PnL}_T^2 / 2\sigma^2}$ at T.
- **Policy** $\pi_\theta(a_t \mid o_t)$: MLP on the current observation (no memory); $V_0 = 0$, $r = 0$, P&L $= V_T - X$.

Remark. (S_t, v_t) is Markov under Heston and both are observed, so in the reported runs o_t carries the full market state: effectively an MDP. Partial observability would arise if v_t were withheld; then a memoryless policy is no longer sufficient.

Backup: from Bellman to actor–critic

Goal: $\max_\pi \mathbb{E}\big[\sum_t \gamma^t r_t\big]$, $\quad Q^\pi(s,a) = \mathbb{E}_\pi\big[\sum_{k\ge 0}\gamma^k r_{t+k} \mid s_t = s,\ a_t = a\big]$.

$$\text{Bellman:}\quad Q^\pi(s,a) = \mathbb{E}\big[r + \gamma\, Q^\pi(s', \pi(s'))\big], \qquad Q^*(s,a) = \mathbb{E}\big[r + \gamma \max_{a'} Q^*(s',a')\big].$$

- **Q-learning** (table): $Q(s,a) \leftarrow Q(s,a) + \eta\,[\,r + \gamma \max_{a'} Q(s',a') - Q(s,a)\,]$; converges to Q^* if every pair is visited infinitely often.
- **DQN**: Q_θ is a neural net, plus replay buffer and target network. Still needs $\max_{a'}$ over a *finite* set of actions.
- **Continuous** $a \in [0,1]$: the max is itself an optimisation, so learn an actor $\mu_\phi(s) \approx \arg\max_a Q(s,a)$ by the deterministic policy gradient $\nabla_\phi J = \mathbb{E}\big[\nabla_a Q(s,a)|_{a=\mu_\phi(s)} \nabla_\phi \mu_\phi(s)\big] \Rightarrow$ DDPG $\Rightarrow$ TD3.
- max over noisy estimates is biased upwards: TD3 uses min of two critics; SAC's soft target $y = r + \gamma\big(\min_i Q_i(s',a') - \beta \log \pi(a' \mid s')\big)$.

Here: $\gamma = 0.99$, MLP actor and critics (Stable-Baselines3 defaults); 400 simulator steps per episode, 22 of them decisions, so the terminal reward is weighted $0.99^{399} \approx 0.02$ at the first decision.

Backup: TD3

Initialise critics $Q_{\theta_1}, Q_{\theta_2}$, actor π_ϕ, targets $\theta_i' \leftarrow \theta_i$, $\phi' \leftarrow \phi$, replay buffer $\mathcal{B}$
for $t = 1$ **to** T **do**
  $a \sim \pi_\phi(s) + \epsilon,\ \epsilon \sim \mathcal{N}(0, \sigma)$; observe r, s'; store (s, a, r, s') in $\mathcal{B}$
  Sample N transitions; $\tilde{a} \leftarrow \pi_{\phi'}(s') + \epsilon',\ \epsilon' \sim \mathrm{clip}(\mathcal{N}(0, \tilde{\sigma}), -c, c)$
  $y \leftarrow r + \gamma \min_{i=1,2} Q_{\theta_i'}(s', \tilde{a})$; update critics by $\min \frac{1}{N} \sum (y - Q_{\theta_i}(s, a))^2$
  if $t \bmod d = 0$ **then**
    $\nabla_\phi J = \frac{1}{N} \sum \nabla_a Q_{\theta_1}(s, a)|_{a=\pi_\phi(s)} \nabla_\phi \pi_\phi(s)$
    $\theta_i' \leftarrow \tau\theta_i + (1-\tau)\theta_i'$, $\phi' \leftarrow \tau\phi + (1-\tau)\phi'$
  end if
end for

Fujimoto, van Hoof & Meger (2018)

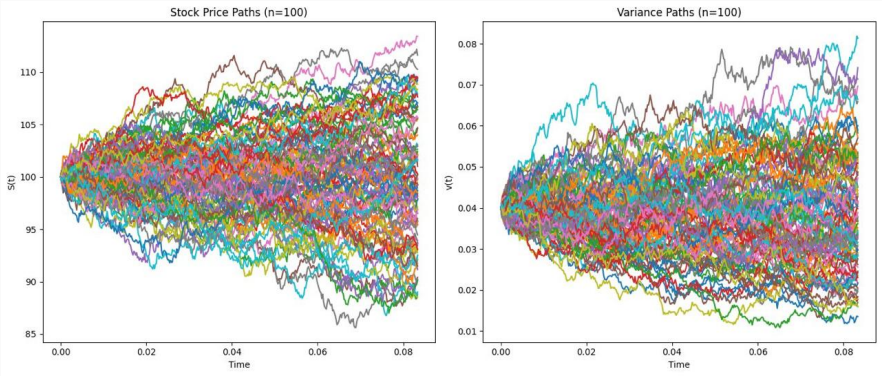
Backup: SAC

Initialise $\psi, \bar{\psi}, \{\theta_i\}_{i=1}^2, \phi$
for each iteration **do**
  for each environment step **do**
    $a_t \sim \pi_\phi(a_t \mid s_t)$, $s_{t+1} \sim p(s_{t+1} \mid s_t, a_t)$, $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, a_t, r_t, s_{t+1})\}$
  end for
  for each gradient step **do**
    $\psi \leftarrow \psi - \lambda_V \nabla_\psi J_V$; $\quad \theta_i \leftarrow \theta_i - \lambda_Q \nabla_{\theta_i} J_Q$; $\quad \phi \leftarrow \phi - \lambda_\pi \nabla_\phi J_\pi$; $\quad \bar{\psi} \leftarrow \tau\psi + (1-\tau)\bar{\psi}$
  end for
end for

Objective: $\max_\pi \ \mathbb{E}\big[\sum_t r_t + \beta\, \mathcal{H}(\pi(\cdot \mid s_t))\big]$, the entropy bonus.

Haarnoja et al. (2018)

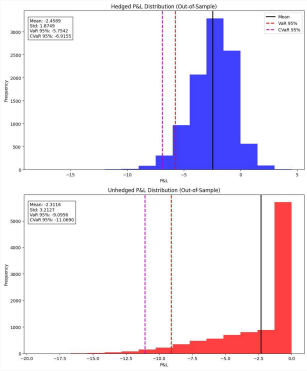
Backup: the Heston simulator



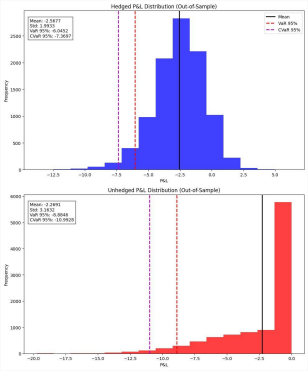
S_0	v_0	θ	σ	κ	ρ	r	T	steps
100	0.04	0.04	0.28	1	-0.7	0	1/12	400

Euler–Maruyama, variance absorbed at 0; trading on 22 of the 400 steps (once a day). Frictionless call price 2.2852.

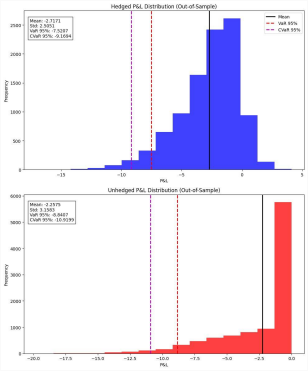
Backup: other runs



SAC, $\alpha = 7, \sigma = 0.5$



SAC, $\alpha = 9, \sigma = 1$



DDPG, $\alpha = 7, \sigma = 0.5$ (unstable)

Backup: risk measures and the delta baseline

For terminal P&L X with CDF F, losses reported as positive numbers:

$$\mathrm{VaR}_\alpha = -\inf\{x : F(x) \geq 1-\alpha\}, \qquad \mathrm{CVaR}_\alpha = -\mathbb{E}\big[X \mid X \leq -\mathrm{VaR}_\alpha\big],$$

with $\alpha = 95\%$. Computed by historical simulation on 10 000 test paths.

Delta baseline. Hold $N(d_1)$ shares with volatility $\sqrt{v_t}$ (the same variance the agent observes), on the same 22 daily dates, positions in $[0, 1]$, same 0.1% costs, same P&L convention.

	Mean	Std	VaR_{95}	CVaR_{95}
No hedge	−2.31	3.22	9.08	11.15
BS delta, 0.1% cost	−2.49	0.50	3.35	3.69
BS delta, no cost	−2.29	0.49	3.12	3.43

Backup: average hedging cost is not a price

Frictionless Heston price of this call: **2.285**.

Mean P&L, starting from zero capital:
TD3 -2.46, BS delta -2.49, BS delta without costs -2.29.

The simulator is risk-neutral with $r = 0$, so for *any* self-financing strategy

$$-\mathbb{E}[\mathrm{PnL}] = \text{price} + \text{expected trading costs}$$

$$2.46 \approx 2.285 + 0.17$$

Reading

2.46 is the average cost of *this* hedge, not a price: the identity holds for every strategy and carries no risk preference. With frictions in an incomplete market there is no unique price; an indifference price needs an explicit utility and real-world dynamics.

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